Routing Evaluation — No Interference

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Contents

1 Evaluation Setup

Identical workloads and scheduler/routing combinations as the `csma_bianchi` interference evaluations, but with `--interference none`: links share bandwidth fairly under concurrent flows but there is no inter-link interference. Results can be compared directly against the interference reports to isolate the cost imposed by wireless contention.

Parameter	**Value**
Interference	**none** (intra-link fair-share only)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Grid routing	W, S, SH, GS, GC, GB, GO, GSD, GSD-D
Random routing	W, S, SH, GO, GS, GSD, GSD-D
Seeds per combo	30
Grid sizes	4×4 (16 nodes, 40 m spacing) and 7×7 (49 nodes)
Random nodes	50, uniform in $[0, L]^2$, comm range $R = 80$ m
Density levels	L150–L500 ($L = 150$ m to 500 m)
DAGs	Small (8 tasks, fork-join), Large (30/60 tasks, pipeline)

Table 1: Evaluation parameters. All other settings identical to the interference reports.

Error bars and uncertainty columns report 95 % confidence intervals ($t_{29} = 2.045$, $n = 30$ seeds): CI $= 2.045 \cdot \sigma/\sqrt{30}$.

2 Grid Network Results

Mean makespan and additional metrics averaged over 30 seeds. **Bold green**: overall experiment winner. Red: worst in that scheduler column.

2.1 4×4 Grid, Small (8 tasks, fork-join)

Routing	**HEFT (s)**	**HEFT-1 (s)**	**HEFT-2 (s)**
W	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
S	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
SH	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GS	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GC	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GB	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GO	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GSD	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GSD-D	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
Best	*W 13.500*±—	*W 13.500*±—	*W 13.500*±—

Table 2: 4×4 small (8 tasks, fork-join) — mean±95 % CI makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
S	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
SH	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GS	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GC	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GB	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GO	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD-D	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0

Table 3: 4×4 small (8 tasks, fork-join) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.2 4×4 Grid, Large (30 tasks, 5-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	130.167±0.000	63.667±0.000	130.167±0.000
S	102.167±0.000	63.667±0.000	102.167±0.000
SH	102.167±0.000	63.667±0.000	102.167±0.000
GS	102.167±0.000	63.667±0.000	102.167±0.000
GC	107.167±0.000	63.667±0.000	107.167±0.000
GB	102.167±0.000	68.333±0.000	102.167±0.000
GO	102.167±0.000	63.667±0.000	102.167±0.000
GSD	70.441±0.000	**45.375**±0.000	70.441±0.000
GSD-D	73.083±0.000	**45.375**±0.000	73.083±0.000
Best	*GSD 70.441*±—	*GSD 45.375*±—	*GSD 70.441*±—

Table 4: 4×4 large (30 tasks, 5-stage pipeline) — mean±95 % CI makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	4.7	0.968	40.8	1.0	0.644	14.6	4.7	0.968	40.8
S	2.0	0.959	32.0	1.0	0.644	14.6	2.0	0.959	32.0
SH	2.0	0.959	32.0	1.0	0.644	14.6	2.0	0.959	32.0
GS	3.2	0.959	32.0	3.3	0.644	14.6	3.2	0.959	32.0
GC	3.2	0.961	36.2	3.6	0.644	14.6	3.2	0.961	36.2
GB	3.9	0.959	32.0	3.4	0.600	15.6	3.9	0.959	32.0
GO	3.2	0.959	32.0	3.3	0.644	14.6	3.2	0.959	32.0
GSD	4.7	0.852	19.4	1.0	0.529	9.5	4.7	0.852	19.4
GSD-D	4.7	0.876	20.0	1.0	0.529	9.5	4.7	0.876	20.0

Table 5: 4×4 large (30 tasks, 5-stage pipeline) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.3 7×7 Grid, Small (8 tasks, fork-join)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
S	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
SH	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GS	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GC	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GB	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GO	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GSD	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
GSD-D	**13.500**±0.000	**13.500**±0.000	**13.500**±0.000
Best	*W 13.500±—*	*W 13.500±—*	*W 13.500±—*

Table 6: 7×7 small (8 tasks, fork-join) — mean±95 % CI makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
S	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
SH	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GS	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GC	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GB	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GO	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0
GSD-D	0.0	0.000	0.0	0.0	0.000	0.0	0.0	0.000	0.0

Table 7: 7×7 small (8 tasks, fork-join) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

2.4 7×7 Grid, Large (60 tasks, 6-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	156.033±2.229	141.357±7.441	155.471±2.096
S	145.100±0.415	142.690±7.320	145.200±0.433
SH	145.333±0.428	134.690±7.527	145.133±0.433
GS	152.071±2.065	147.193±6.387	153.921±2.100
GC	152.908±2.408	142.157±7.069	152.854±2.282
GB	152.358±2.313	142.690±7.320	154.438±2.092
GO	154.058±1.807	145.357±6.961	153.854±2.079
GSD	130.041±12.346	**71.483**±0.707	128.321±13.262
GSD-D	122.935±13.365	72.417±0.734	136.970±10.655
Best	*GSD-D 122.935±—*	*GSD 71.483±—*	*GSD 128.321±—*

Table 8: 7×7 large (60 tasks, 6-stage pipeline) — mean±95 % CI makespan (s) over 30 seeds.

Routing	HEFT			HEFT-1			HEFT-2		
	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)	Hops	P-LU	XD(s)
W	6.0	0.930	30.2	1.0	0.933	27.7	5.9	0.931	30.0
S	3.9	0.936	25.2	1.0	0.933	28.4	3.8	0.937	25.3
SH	4.0	0.937	25.3	1.0	0.929	25.0	4.0	0.936	25.2
GS	5.8	0.937	28.1	2.4	0.927	31.5	5.9	0.936	29.0
GC	6.4	0.934	28.6	2.7	0.933	27.8	6.2	0.936	28.5
GB	6.3	0.936	28.3	2.3	0.933	27.3	6.2	0.935	29.3
GO	5.9	0.936	29.0	2.4	0.935	29.0	5.9	0.936	28.9
GSD	5.5	0.862	23.0	1.0	0.786	13.7	5.6	0.847	23.0
GSD-D	5.4	0.831	21.5	1.0	0.773	13.8	5.9	0.885	24.3

Table 9: 7×7 large (60 tasks, 6-stage pipeline) — additional metrics averaged over 30 seeds. Hops = mean path length; P-LU = peak link utilisation; XD = mean transfer duration.

3 Random Network Results

3.1 Topology Statistics

Level	Side L (m)	Links	Avg. degree
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 10: Random network topology statistics (seed 42).

3.2 Makespan vs. Density

Each point is the best-routing mean makespan $\pm$ 95 % CI ($t_{29} = 2.045$) over 30 seeds.

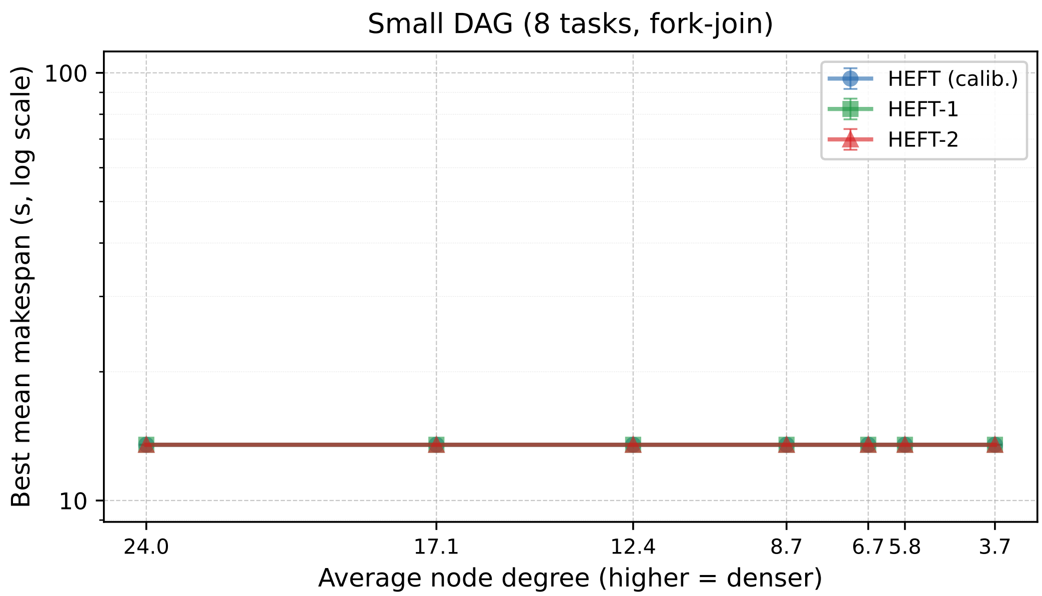

Figure 1: No-interference: best-routing mean makespan vs. avg node degree — Small DAG (8 tasks, fork-join). Error bars show 95 % CI over 30 seeds.

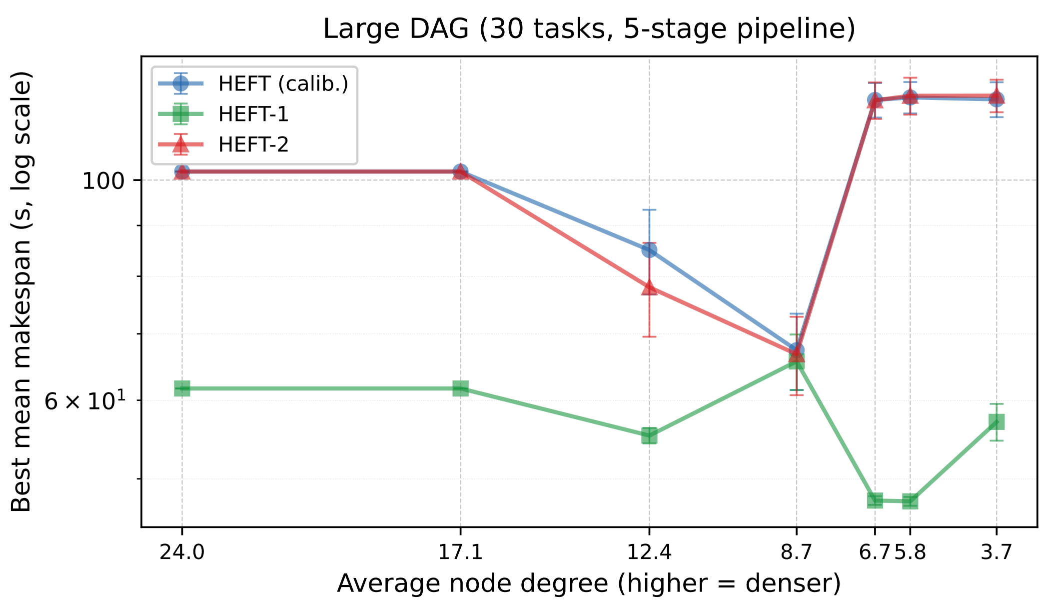

Figure 2: No-interference: best-routing mean makespan vs. avg node degree — Large DAG (30 tasks, 5-stage pipeline). Error bars show 95 % CI over 30 seeds.

3.3 Additional Metrics vs. Density (Large DAG)

Metrics are taken at the *best* routing scheme for each scheduler at each density level, averaged over 30 seeds.

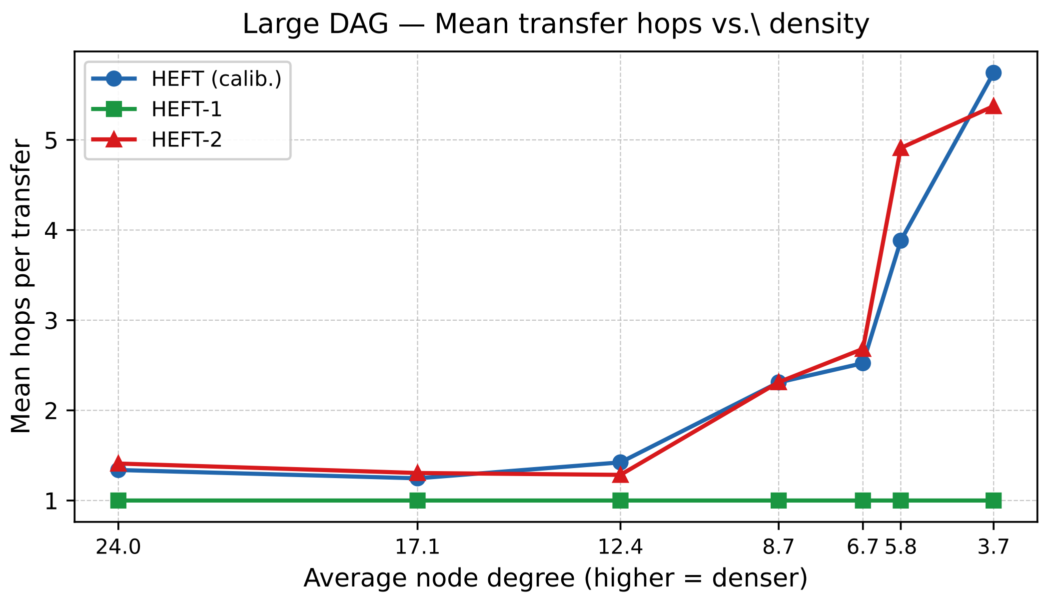

Figure 3: Mean hops per transfer for the best routing scheme at each density.

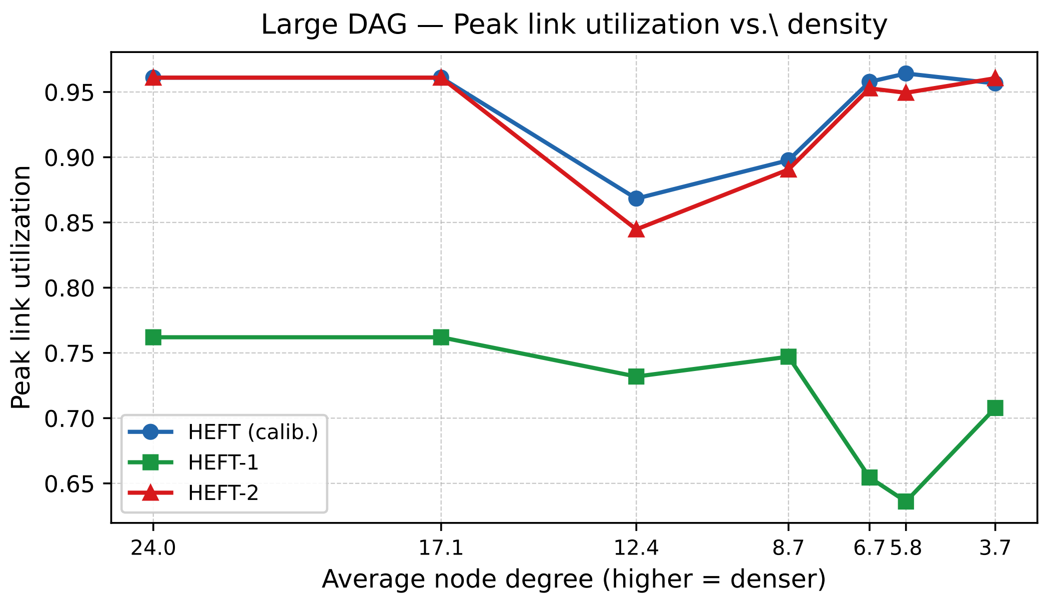

Figure 4: Peak link utilisation (max over all links) for the best routing scheme.

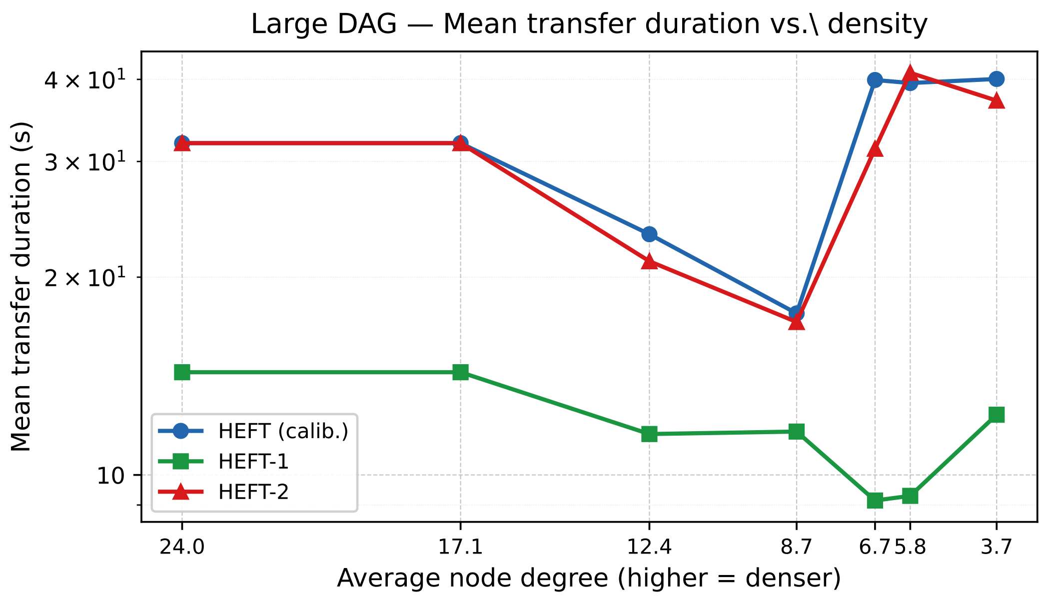

Figure 5: Mean transfer duration (s, log scale) for the best routing scheme.

3.4 Best Routing Summary — Small DAG (8 tasks)

Density	Deg.	HEFT			HEFT-1			HEFT-2		
		Rt	ms(s)	H	Rt	ms(s)	H	Rt	ms(s)	H
L150	24.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L200	17.1	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L250	12.4	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L300	8.7	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L350	6.7	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L400	5.8	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0
L500	3.7	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0	W	**13.500**±0.000	0.0

Table 11: Small DAG (8 tasks) — best routing per (density, scheduler). ms = mean±95 % CI; H = mean hops. **Bold green**: overall best.

3.5 Best Routing Summary — Large DAG (30 tasks)

Density	Deg.	HEFT			HEFT-1			HEFT-2		
		Rt	ms(s)	H	Rt	ms(s)	H	Rt	ms(s)	H
L150	24.0	W	102.000±0.000	1.3	W	61.667±0.000	1.0	W	102.000±0.000	1.4
L200	17.1	W	102.000±0.000	1.2	W	61.667±0.000	1.0	W	102.000±0.000	1.3
L250	12.4	GSD-D	85.017±8.323	1.4	GSD-D	55.278±0.992	1.0	GSD-D	77.983±8.457	1.3
L300	8.7	GSD-D	67.394±5.958	2.3	GSD	65.667±4.227	1.0	GSD-D	66.767±6.055	2.3
L350	6.7	S	120.433±4.821	2.5	GSD	47.544±0.477	1.0	GSD-D	120.333±5.098	2.7
L400	5.8	S	121.133±4.407	3.9	GSD-D	**47.458**±0.485	1.0	W	121.600±5.206	4.9
L500	3.7	S	120.600±4.881	5.7	GSD-D	57.061±2.433	1.0	GSD-D	121.633±4.585	5.4

Table 12: Large DAG (30 tasks) — best routing per (density, scheduler). ms = mean±95 % CI; H = mean hops. **Bold green**: overall best.

4 Full Random Network Results

Mean±95 % CI makespan (s), mean hops, and peak link util over 30 seeds. **Bold green**: overall best for that density+DAG. Red: worst in column.

4.1 L150 (avg. degree 24.0) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 13: L150 ($L = 150\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.2 L200 (avg. degree 17.1) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 14: L200 ($L = 200$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.3 L250 (avg. degree 12.4) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 15: L250 ($L = 250$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.4 L300 (avg. degree 8.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 16: L300 ($L = 300$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.5 L350 (avg. degree 6.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 17: L350 ($L = 350$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.6 L400 (avg. degree 5.8) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 18: L400 ($L = 400$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.7 L500 (avg. degree 3.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
S	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
SH	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GO	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GS	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000
GSD-D	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000	**13.500**±0.000	0.0	0.000

Table 19: L500 ($L = 500$ m) — Small DAG (8 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.8 L150 (avg. degree 24.0) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	102.000±0.000	1.3	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.4	0.961
S	102.000±0.000	1.3	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.3	0.961
SH	102.000±0.000	1.4	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.2	0.961
GO	107.000±0.000	4.0	0.963	**61.667**±0.000	4.0	0.762	107.000±0.000	4.0	0.963
GS	107.000±0.000	4.0	0.963	**61.667**±0.000	4.0	0.762	107.000±0.000	4.0	0.963
GSD	107.000±0.000	1.3	0.963	**61.667**±0.000	1.0	0.762	107.000±0.000	1.4	0.963
GSD-D	107.000±0.000	1.3	0.963	**61.667**±0.000	1.0	0.762	107.000±0.000	1.2	0.963

Table 20: L150 ($L = 150\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.9 L200 (avg. degree 17.1) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	102.000±0.000	1.2	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.3	0.961
S	102.000±0.000	1.3	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.3	0.961
SH	102.000±0.000	1.4	0.961	**61.667**±0.000	1.0	0.762	102.000±0.000	1.4	0.961
GO	107.000±0.000	4.0	0.963	**61.667**±0.000	4.0	0.762	107.000±0.000	4.0	0.963
GS	107.000±0.000	4.0	0.963	**61.667**±0.000	4.0	0.762	107.000±0.000	4.0	0.963
GSD	107.000±0.000	1.4	0.963	**61.667**±0.000	1.0	0.762	107.000±0.000	1.3	0.963
GSD-D	107.000±0.000	1.3	0.963	**61.667**±0.000	1.0	0.762	107.000±0.000	1.3	0.963

Table 21: L200 ($L = 200\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.10 L250 (avg. degree 12.4) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	122.300±4.801	1.3	0.956	61.667±0.000	1.0	0.762	125.100±4.568	1.4	0.953
S	124.267±4.357	1.3	0.959	61.667±0.000	1.0	0.762	122.967±4.994	1.4	0.951
SH	127.467±4.349	1.5	0.947	61.667±0.000	1.0	0.762	128.000±3.669	1.4	0.955
GO	105.000±0.930	4.0	0.962	61.667±0.000	4.0	0.762	103.833±0.915	4.0	0.962
GS	105.000±0.930	4.0	0.962	61.667±0.000	4.0	0.762	103.667±0.895	4.0	0.962
GSD	88.300±7.906	1.3	0.882	55.722±0.866	1.0	0.758	80.083±8.802	1.3	0.847
GSD-D	85.017±8.323	1.4	0.868	**55.278**±0.992	1.0	0.732	77.983±8.457	1.3	0.845

Table 22: L250 ($L = 250\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.11 L300 (avg. degree 8.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	127.900±4.053	2.4	0.950	71.828±7.439	1.0	0.771	123.067±4.703	2.3	0.956
S	121.867±4.984	2.3	0.953	68.037±6.820	1.0	0.751	122.133±4.749	2.3	0.957
SH	128.500±3.751	2.4	0.951	77.074±7.720	1.0	0.800	124.933±4.528	2.4	0.954
GO	102.833±0.708	3.8	0.961	72.382±8.433	3.1	0.775	103.000±0.759	3.8	0.961
GS	102.500±0.570	3.8	0.961	71.838±6.305	2.7	0.815	102.500±0.570	3.7	0.961
GSD	67.472±6.607	2.4	0.894	**65.667**±4.227	1.0	0.747	68.117±6.513	2.3	0.901
GSD-D	67.394±5.958	2.3	0.898	66.521±4.218	1.0	0.755	66.767±6.055	2.3	0.891

Table 23: L300 ($L = 300$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.12 L350 (avg. degree 6.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	128.000±3.669	2.7	0.955	61.578±0.656	1.0	0.705	121.200±4.768	2.5	0.958
S	120.433±4.821	2.5	0.958	62.114±0.551	1.0	0.686	122.300±4.801	2.6	0.956
SH	124.167±4.674	2.7	0.954	62.067±0.469	1.0	0.712	126.800±4.229	2.8	0.952
GO	123.567±4.842	3.0	0.952	60.430±0.738	2.4	0.715	125.267±4.606	3.1	0.952
GS	121.200±4.768	2.9	0.958	61.285±0.702	2.7	0.708	125.100±4.568	3.1	0.953
GSD	123.067±4.703	2.6	0.956	**47.544**±0.477	1.0	0.654	121.967±4.696	2.5	0.958
GSD-D	122.467±4.851	2.7	0.955	47.718±0.454	1.0	0.643	120.333±5.098	2.7	0.953

Table 24: L350 ($L = 350$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.13 L400 (avg. degree 5.8) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	126.467±4.162	4.7	0.954	60.967±0.772	1.0	0.696	121.600±5.206	4.9	0.949
S	121.133±4.407	3.9	0.964	61.456±0.687	1.0	0.703	122.733±4.604	4.4	0.958
SH	123.400±4.797	4.5	0.954	61.138±0.717	1.0	0.710	124.933±4.528	5.1	0.954
GO	125.867±4.396	4.9	0.953	60.942±0.765	2.3	0.699	125.700±4.359	4.9	0.954
GS	124.333±4.716	5.2	0.953	61.602±0.661	2.7	0.702	126.467±4.162	4.8	0.954
GSD	125.367±4.281	4.7	0.957	47.544±0.477	1.0	0.630	123.067±4.703	4.6	0.956
GSD-D	122.300±4.801	4.6	0.956	**47.458**±0.485	1.0	0.636	125.700±4.359	4.8	0.954

Table 25: L400 ($L = 400$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

4.14 L500 (avg. degree 3.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms(s)	H	PLU	ms(s)	H	PLU	ms(s)	H	PLU
W	122.133±4.749	5.9	0.957	61.194±0.751	1.0	0.719	124.933±4.528	6.3	0.954
S	120.600±4.881	5.7	0.957	60.833±0.906	1.0	0.691	124.600±4.445	5.9	0.956
SH	121.200±4.768	5.7	0.958	60.683±0.853	1.0	0.702	129.000±3.823	6.7	0.948
GO	124.500±4.757	6.5	0.951	61.100±0.835	1.7	0.704	126.033±4.431	6.0	0.952
GS	124.600±4.445	6.0	0.956	61.567±0.799	1.7	0.702	126.733±3.831	5.6	0.958
GSD	123.833±4.585	5.5	0.956	58.050±2.002	1.0	0.701	124.933±4.528	5.9	0.954
GSD-D	125.533±4.321	6.1	0.955	**57.061**±2.433	1.0	0.708	121.633±4.585	5.4	0.961

Table 26: L500 ($L = 500$ m) — Large DAG (30 tasks). ms = mean±95 % CI; H = mean hops; PLU = peak link util.

5 Analysis

5.1 Small DAG: Compute-Bound, All Metrics Trivial

All routing schemes and schedulers produce identical makespan (13.500 s) on the small DAG across both grid sizes and all density levels. Transfers are same-node (0 hops, 0 link utilisation) because HEFT-1 co-locates all 8 tasks. Even under HEFT/HEFT-2, the DAG's compute time dominates: transfer costs are negligible without interference.

5.2 Why GSD/GSD-D Win on the Large DAG

The extra metrics reveal the mechanism clearly. On the 7×7 grid large DAG under HEFT-1:

Routing	Makespan (s)	Mean hops	Peak link util.	Mean xfer dur. (s)
W	141.357	1.0	0.933	27.7
S	142.690	1.0	0.933	28.4
SH	134.690	1.0	0.929	25.0
GS	147.193	2.4	0.927	31.5
GO	145.357	2.4	0.935	29.0
GSD	71.483	1.0	0.786	13.7
GSD-D	72.417	1.0	0.773	13.8

W, S, and SH all use 1-hop paths but achieve peak link utilisation of 0.93, with mean transfer durations of 25–28 s. GS and GO use 2.4–2.7 hops on average. GSD and GSD-D also use 1-hop paths but reduce peak link utilisation to 0.77–0.79 and halve the mean transfer duration to 13.7 s, directly explaining the 2× makespan advantage.

5.3 Comparison to csma_bianchi Results

Removing interference collapses the large-DAG makespan from 150–927 s (csma_bianchi, best routing) to 47–66 s (no interference, best routing) under HEFT-1—a 3–14× reduction. This confirms that the dominant cost in the csma_bianchi experiments was inter-link spectrum contention, not intra-link queuing.

5.4 Reproducing These Results

```
cd ncsim/
python run_no_interference_eval.py      # ~70 min (12 060 runs)
python compute_noint_augmented.py       # reads existing traces
python gen_no_interference_report.py    # regenerates tex + plots
cd docs/ && pdflatex no_interference_results.tex
```